

Chomsky Hierarchy

In 1957, Noam Chomsky published Syntactic Structures called as Chomsky Hierarchy.

Grammar Type	Grammar Name	Language Generated	Automaton To Accept	Productions	Example
Type 0	Unrestricted	Recursively Enumerable	Turing Machine	No Restrictions $\alpha \rightarrow \beta$ where $\alpha \neq \epsilon$	$A \rightarrow AB$ $AB \rightarrow BC$ $B \rightarrow AcD$ $Ac \rightarrow abc$ $D \rightarrow \epsilon$
Type 1	Context Sensitive	Context Sensitive	Linear Bounded Automaton	$\alpha_1 A \alpha_2 \rightarrow \alpha_1 \beta \alpha_2$, $\beta \neq \epsilon$ $\alpha \rightarrow \beta$ len of β is at least len of α , except $S \rightarrow \epsilon$ is allowed in SS	$A \rightarrow AB$ $AB \rightarrow AC$ $AC \rightarrow ab$
Type 2	Context Free	Context Free	Pushdown Automaton	$A \rightarrow \alpha$	$S \rightarrow aSa bSb a b$
Type 3	Regular(Left & Right Linear)	Regular	Finite State Machine	$A \rightarrow a$ and $A \rightarrow aB$	<i>Left Linear:</i> $S \rightarrow Ca Bb$ $C \rightarrow Bb$ $B \rightarrow Ba b$ <i>Right Linear:</i> $S \rightarrow 0S$ $A \rightarrow 0A 1$

Where, A,B are Non terminals, a,b are Terminals and α, β, γ are sentential forms.

Types of Grammar

On the basis of types of production rules

- Type 0 – Unrestricted Grammar
- Type 1 - Context Sensitive Grammar
- Type 2 - Context Free Grammar
- Type 3 – Regular Grammar (Left & Right Linear)

On the basis of number of derivation trees

- Ambiguous
- Unambiguous

On the basis of Number of Strings in a Language

- Recursive Grammar
- Non Recursive Grammar